Yihan Wu

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AI/ML Systems | Software Engineering | Signal Processing & Edge AI

EDUCATION

Columbia University New York, NY

M.S. in Electrical Engineering, AI Engineering Specialization *Expected Dec 2027*

Coursework: Designing LLM Agents; Systems & Network for Large-Scale LLM Inference; Heterogeneous Computing; Big Data Analytics

University of Wisconsin-Madison Madison, WI

B.S. in Data Science and Neurobiology; Certificate: Mathematics *Aug 2026*

Coursework: Artificial Intelligence; Numerical Linear Algebra; Probability; Signals, Information & Computation

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL

Frameworks: PyTorch, TensorFlow, Keras, LangChain, vLLM

AI/ML: machine learning, deep learning, neural networks, LLM agents, agentic AI, statistical modeling

Systems & MLOps: Docker, Kubernetes, Linux, Git, Spark, Hadoop, Apache Airflow, BigQuery, CUDA, OpenCL

Engineering: real-time systems, signal processing, backend development, software testing, edge AI, agentic workflow

EXPERIENCE

Developer Apr 2026 - Aug 2026

UW-Madison College of Computing & Artificial Intelligence *Madison, WI*

• Developed and optimized a real-time audio-to-score alignment backend, adapting an existing open-source score-following toolkit with streaming audio features and Online Time Warping for live-performance tracking.

• Collaborated with frontend and research teams on backend-frontend integration and mobile testing.

• Delivered a robust backend for an accessibility-focused HCI project whose demo paper was accepted to ACM UIST 2026.

Developer Aug 2024 - May 2026

UW-Madison School of Medicine and Public Health *Madison, WI*

• Engineered an end-to-end MATLAB pipeline that automated EEG source reconstruction, reducing a months-long workflow to under 2 days while processing approximately 30,000 epochs.

• Developed a Python framework for time-series processing, hypothesis testing, and logistic regression-based EEG decoding.

• Managed Git repositories and technical documentation; standardized outputs and coordinated deliverables across collaborators.

PROJECTS & LEADERSHIP

NPU-Piggy - AI Infrastructure Lead, MadData Qualcomm Track Feb 2026

• Selected 1 of 10 Qualcomm-track teams from 191 applicants; deployed EasyOCR through Qualcomm AI Hub with NPU runtime, accelerating receipt processing by approximately 60%.

• Integrated Mistral 7B, LangChain, and Pydantic to convert noisy OCR into validated JSON for a React dashboard with zero cloud dependency.

YATA VioLin - Lead Developer, MadHacks Nov 2025

• Engineered a PyTorch-based pipeline that transcribed violin audio to MIDI and compared reference/student performances using custom DTW and greedy note matching.

• Detected pitch/timing error patches and generated visual, teacher-style feedback with Gradio and Gemini 2.0 through LangChain.

Founder & President, WISA Jan 2023 - Sep 2024

• Founded and grew WISA to 300+ members; secured a campus grant and led the executive board and multicultural events programming.